Investment Monte Carlo Projection - Data Flow Documentation

This document maps the complete data flow for Monte Carlo projections, using the Mitchell persona (peak_earners) as a working example.

Mitchell Portfolio Overview

Investment Accounts

Account	Owner	Type	Value	Risk Preference	Custom Risk
David's S&S ISA	David	ISA	£95,000	high	Yes
Sarah's S&S ISA	Sarah	ISA	£85,000	medium	No
Joint GIA	Joint	GIA	£95,000	upper_medium	No
VCT Holdings	David	VCT	£30,000	high	Yes

DC Pensions (also use Monte Carlo)

Account	Owner	Type	Value	Risk Preference
Global Finance Corp Pension	David	Workplace	£180,000	upper_medium
David's SIPP	David	SIPP	£320,000	upper_medium

User Risk Profiles

User	Main Risk Level	Source
David	upper_medium	Self-assessed
Sarah	medium	Self-assessed

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1. Portfolio-Level Monte Carlo

Data Sources

File: `app/Services/Investment/InvestmentProjectionService.php` **Method**: `getPortfolioProjections()` (lines 51-94)

Step 1: Calculate Total Portfolio Value

```
Portfolio Value = Σ (Account Value × Ownership Percentage / 100)
```

For Mitchells (David's view):

```
David's S&S ISA:    £95,000 × 100% = £95,000
Sarah's S&S ISA:    £85,000 × 0%   = £0 (spouse-owned)
Joint GIA:          £95,000 × 50%  = £47,500
VCT Holdings:       £30,000 × 100% = £30,000
─────────────────────────────────────────────
Total Portfolio Value:               £172,500
```

Source: `InvestmentAccount` model fields:

- `current_value` (float)
- `ownership_type` (individual/joint/trust)
- `ownership_percentage` (decimal)

Step 2: Calculate Weighted Average Risk

File: `app/Services/Investment/InvestmentProjectionService.php` **Method**: `calculateWeightedPortfolioRisk()` (lines 322-368)

```
Risk Levels → Numeric Values:
  low           = 1
  lower_medium  = 2
  medium        = 3
  upper_medium  = 4
  high          = 5
```

For Mitchells (David's view):

```
Account                 | Value    | Weight | Risk       | Numeric | Weighted
────────────────────────|──────────|────────|────────────|─────────|─────────
David's S&S ISA         | £95,000  | 55.1%  | high       | 5       | 2.755
Joint GIA (50%)         | £47,500  | 27.5%  | upper_med  | 4       | 1.100
VCT Holdings            | £30,000  | 17.4%  | high       | 5       | 0.870
────────────────────────|──────────|────────|────────────|─────────|─────────
Total                   | £172,500 | 100%   |            |         | 4.725

Weighted Risk = 4.725 → Rounds to "upper_medium" (4-5 range)
```

Source: Each account's risk determined by:

1. `risk_preference` field (if `has_custom_risk = true`)
2. User's main `risk_level` from `RiskProfile` model
3. Default: "medium"

Step 3: Get Return Parameters for Portfolio Risk

File: `app/Services/Risk/RiskPreferenceService.php` **Method**: `getReturnParameters()` (lines 314-324)

Risk Level	Expected Return (Typical)	Volatility
low	2.0%	3%
lower_medium	3.5%	6%
medium	5.0%	10%
upper_medium	6.5%	15%
high	8.0%	20%

For Mitchells' portfolio (upper_medium):

- **Expected Return**: 6.5%
- **Volatility**: 15%

Step 4: Estimate Portfolio Contributions

File: `app/Services/Investment/ContributionEstimatorService.php`

Account Type	Estimation Method
ISA	`isa_subscription_current_year ÷ months_elapsed` OR `£20,000 ÷ 12`
GIA	`current_value × 5% ÷ 12`
VCT	£0 (typically no regular contributions)

For Mitchells:

```
David's S&S ISA:   £10,000 ÷ 10 months = £1,000/month
Joint GIA (50%):   £47,500 × 5% ÷ 12   = £198/month
VCT Holdings:      £0/month
```

```
────────────────────────────────────────────
Estimated Monthly: £1,198/month
```

Step 5: Run Monte Carlo Simulation

File: `app/Services/Investment/MonteCarloSimulator.php` **Method**: `simulate()` (lines 20-94)

Input Parameters:

```
$simulator->simulate(
    startValue: 172500,           // Portfolio value
    monthlyContribution: 1198,    // Estimated contributions
    expectedReturn: 0.065,        // 6.5% annual
    volatility: 0.15,             // 15% annual
    years: 20,                    // Projection period
    iterations: 1000              // Simulation runs
);
```

Algorithm:

1. Convert annual → monthly: `monthlyReturn = 0.065/12`, `monthlyVol = 0.15/√12`
2. For each of 1,000 iterations:
   - For each month:
     - Generate random return using Box-Muller normal distribution
     - `portfolio = portfolio × (1 + randomReturn) + monthlyContribution`
   - Store year-end values
3. Calculate percentiles across all iterations

Output: Year-by-year percentile bands (p5, p10, p15, p20, p50, p75, p90)

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2. Individual Account Monte Carlo

Data Sources per Account

File: `app/Services/Investment/InvestmentProjectionService.php` **Method**: `calculateAccountProjection()` (lines 156-226)

Risk Priority Hierarchy

```
1. Account has risk_preference AND has_custom_risk = true
   → Use account's risk_preference

2. Account has risk_preference but has_custom_risk = false
   → Use user's main risk_level

3. Account has no risk_preference
   → Use user's main risk_level

4. User has no risk profile
   → Default to "medium"
```

Mitchell Individual Account Projections

David's S&S ISA:

```
Value:           £95,000 × 100% = £95,000
Risk Source:     Account-level (has_custom_risk = true)
Risk Level:      high
Expected Return: 8.0%
Volatility:      20%
Est. Contribution: £1,000/month (from ISA subscription)
```

Sarah's S&S ISA (spouse account, not in David's portfolio):

```
Value:           £85,000 × 100% = £85,000
Risk Source:     Account-level
Risk Level:      medium
Expected Return: 5.0%
Volatility:      10%
Est. Contribution: £833/month (£10,000 ÷ 12)
```

Joint GIA (David's 50% share):

```
Value:           £95,000 × 50% = £47,500
Risk Source:     Account-level
Risk Level:      upper_medium
Expected Return: 6.5%
Volatility:      15%
Est. Contribution: £198/month (5% of share ÷ 12)
```

VCT Holdings:

```
Value:           £30,000 × 100% = £30,000
Risk Source:     Account-level (has_custom_risk = true)
Risk Level:      high
Expected Return: 8.0%
Volatility:      20%
Est. Contribution: £0/month
```

Monte Carlo per Account

Each account runs independent simulation with its own parameters:

```php
// David's S&S ISA
$simulator->simulate(95000, 1000, 0.08, 0.20, 20, 1000);

// Joint GIA (David's share)
$simulator->simulate(47500, 198, 0.065, 0.15, 20, 1000);

// VCT Holdings
$simulator->simulate(30000, 0, 0.08, 0.20, 20, 1000);
```

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3. Risk Assignment Flow

How Risk Gets to Each Account

```
┌─────────────────────────────────────────────────────────────────────┐
│                      RISK ASSIGNMENT FLOW                           │
├─────────────────────────────────────────────────────────────────────┤
│                                                                     │
│                                                                     │
│  ┌──────────────────┐                                               │
│  │ RiskProfile Model│  ← User completes risk questionnaire          │
│  │ (per user)       │                                               │
│  │                  │                                               │
│  │ • risk_level     │  ← "low" / "lower_medium" / "medium" /       │
│  │ • is_self_assess │    "upper_medium" / "high"                   │
│  └──────────────────┘                                               │
│           │                                                         │
│           ▼                                                         │
│  ┌────────────────────────────────────────────────────────────┐     │
│  │              InvestmentAccount Model                        │       │
```

```
│   │                                                    │      │
│   │  risk_preference ──────┬── If has_custom_risk = true │      │
│   │                        │   → Use this value          │      │
│   │                        │                             │      │
│   │  has_custom_risk ──────┼── If false or not set       │      │
│   │                        │   → Fall back to user's main│      │
│   │                        │     risk_level              │      │
│   │                        │                             │      │
│   │                        └── If no user risk profile   │      │
│   │                            → Default to "medium"     │      │
│   └──────────────────────────────────────────────────────┘      │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

Risk Constraints

File: `app/Services/Risk/RiskPreferenceService.php` **Method**: `getAllowedProductRiskLevels()` (lines 216-236)

Account-level risk must be within ±1 of user's main risk level:

User Main Risk	Allowed Account Risks
low	low, lower_medium
lower_medium	low, lower_medium, medium
medium	lower_medium, medium, upper_medium
upper_medium	medium, upper_medium, high
high	upper_medium, high

For David (main: upper_medium):

- Can set accounts to: medium, upper_medium, or high ✓
- His "high" ISA is valid (within ±1)

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4. Data Flow Diagram

```
┌────────────────────────────────────────────────────────────────────────┐
│                         MONTE CARLO DATA FLOW                          │
├────────────────────────────────────────────────────────────────────────┤
│                                                                        │
│  DATABASE                  SERVICES                   FRONTEND         │
│  ────────                  ────────                   ────────         │
│                                                                        │
│  ┌─────────────────┐                                                   │
│  │ investment_     │                                                   │
│  │ accounts        │───┐                                               │
│  │                 │   │                                               │
│  │ • current_value │   │    ┌──────────────────────┐                   │
│  │ • ownership_%   │   ├───►│ InvestmentProjection │                   │
│  │ • risk_pref     │   │    │ Service              │                   │
│  │ • has_custom    │   │    │                      │                   │
│  └─────────────────┘   │    │ • getPortfolio       │                   │
│                        │    │   Projections()      │                   │
│  ┌─────────────────┐   │    │                      │    ┌─────────────┐│
│  │ risk_profiles   │───┤    │ • calculateAccount   │───►│ API Response││
│  │                 │   │    │   Projection()       │    │             ││
│  │ • risk_level    │   │    │                      │    │ • percentile││
│  │ • user_id       │   │    │ • calculateWeighted  │    │   bands     ││
│  └─────────────────┘   │    │   PortfolioRisk()    │    │ • year_by_  ││
│                        │    └──────────────────────┘    │   year      ││
│  ┌─────────────────┐   │               │                │ • risk_level││
│  │ users           │───┘               │                └─────────────┘│
│  │                 │                   ▼                       │       │
│  │ • id            │         ┌────────────────────────┐        │       │
│  │ • spouse_id     │         │ RiskPreferenceService  │        │       │
```

```
│  └─────────────────┘         │                        │           │         │
│                              │ • getReturnParameters()│           │         │
│                              │   → expected_return    │           ▼         │
│                              │   → volatility         │  ┌────────────────┐ │
│                              └────────────────────────┘  │ Vue Component  │ │
│                                           │              │                │ │
│                                           ▼              │ Investment     │ │
│                              ┌────────────────────────┐  │ Projection     │ │
│                              │ MonteCarloSimulator    │  │ Chart.vue      │ │
│                              │                        │  │                │ │
│                              │ • simulate()           │  │ • ApexCharts   │ │
│                              │   - 1000 iterations    │  │ • Area chart   │ │
│                              │   - Box-Muller random  │  │ • Percentile   │ │
│                              │   - Year-end captures  │  │   bands        │ │
│                              │                        │  └────────────────┘ │
│                              │ • calculatePercentiles │                     │
│                              │   → p5, p10, p15, p20  │                     │
│                              │   → p50, p75, p90      │                     │
│                              └────────────────────────┘                     │
│                                                                             │
└─────────────────────────────────────────────────────────────────────────────┘
```

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5. API Endpoints

Portfolio Projections

```
POST /api/investment/projections

Request:
{
  "projection_periods": [5, 10, 20, 30],
  "selected_period": 20,
  "contribution_overrides": {
    "1": 1200,  // Override account ID 1 contribution
    "3": 500    // Override account ID 3 contribution
  }
}

Response:
{
  "portfolio": {
    "total_value": 172500,
    "risk_level": "upper_medium",
    "expected_return": 6.5,
    "volatility": 15,
    "projections": {
      "20": {
        "years": 20,
        "median_value": 892450,
        "percentiles": {...},
        "year_by_year": [...]
      }
    }
  },
  "accounts": [
    {
      "account_id": 1,
      "account_name": "David's S&S ISA",
      "risk_level": "high",
      "projections": {...}
    },
    ...
  ]
}
```

Individual Account Projections (with risk override)

```
GET /api/investment/accounts/{id}/projections?risk_level=medium

Response:
{
  "account_id": 1,
  "account_name": "David's S&S ISA",
  "risk_level": "medium",        // Override applied
  "risk_source": "override",
  "expected_return": 5.0,
  "volatility": 10,
  "projections": {...}
}
```

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6. Key Files Reference

Component	File	Key Methods
Monte Carlo Engine	`app/Services/Investment/MonteCarloSimulator.php`	`simulate()`, `calculatePercentiles()`
Portfolio Projections	`app/Services/Investment/InvestmentProjectionService.php`	`getPortfolioProjections()`, `calculateAccountProjection()`
Risk Parameters	`app/Services/Risk/RiskPreferenceService.php`	`getReturnParameters()`, `getMainRiskLevel()`
Contribution Estimation	`app/Services/Investment/ContributionEstimatorService.php`	`estimateAccountContribution()`
Controller	`app/Http/Controllers/Api/InvestmentController.php`	`getAccountProjections()`, `startMonteCarlo()`
Frontend Chart	`resources/js/components/Investment/InvestmentProjectionChart.vue`	ApexCharts area chart
Results Display	`resources/js/components/Investment/MonteCarloResults.vue`	Full results modal

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7. Risk Level Parameters Summary

Risk Level	Numeric	Expected Return	Volatility	Equity %	Bond %
low	1	2.0%	3%	10%	70%
lower_medium	2	3.5%	6%	30%	55%
medium	3	5.0%	10%	50%	40%
upper_medium	4	6.5%	15%	75%	20%
high	5	8.0%	20%	90%	5%

---

8. Mitchell Example: 20-Year Portfolio Projection

Input:

- Start Value: £172,500
- Monthly Contribution: £1,198
- Expected Return: 6.5%
- Volatility: 15%
- Years: 20
- Iterations: 1,000

Output (example percentiles at year 20):

- 10th percentile: £485,000 (poor market scenario)

- 25th percentile: £620,000
- 50th percentile (median): £892,000
- 75th percentile: £1,280,000
- 90th percentile: £1,850,000 (strong market scenario)

Total contributions over 20 years: £172,500 + (£1,198 × 240) = £460,020 Median growth: £892,000 - £460,020 = £431,980 (growth from returns)